RADIO NAVIGATION RECEIVING SET, OMEGA,
AN/BRN-7, AN/SRN-17

This amendment forms a part of Military Specification MIL-R-28819(EC), dated 19 May 1975, and is approved for use by the Naval Electronic Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

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2.1, "SPECIFICATIONS", "MILITARY", after "MIL-S-901", Add:

"MIL-C-915/6 — Cable, Electrical, 600 Volts, Types SHOF, DHOF, THOF, And FHOF".

2.1, "SPECIFICATIONS", "MILITARY", after "MIL-E-17555", Add:

"MIL-T-28820 — Radio Navigation Receiving Set, Omega, AN/BRN-7, AN/SRN-7".

2.1: "STANDARDS", "MILITARY": Delete "MIL-STD-167 Mechanical Vibrations of Shipboard Equipment" and substitute "MIL-STD-167-1 Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)".

2.1, "STANDARDS", "MILITARY", after "MIL-STD-167-1", Add:

"MIL-STD-454 — Standard General Requirements For Electronic Equipment".

2.1 "STANDARDS", "MILITARY": Delete "MIL-STD-480 Configuration Control-Engineering Changes, Deviations and Waivers".

Bottom of page, Add:

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter."

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2.1 "STANDARDS", "MILITARY": "MIL-STD-781": Delete "Reliability Tests Exponential Distribution" and substitute "Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution".

2.1 "STANDARDS", "MILITARY", after "MIL-STD-781: Delete "MIL-STD-1399 - Interface Standard For Shipboard Systems" and substitute "MIL-STD-1399, Section 702 - Interface Standard For Shipboard Systems Synchro Data Transmission".

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3.3.3: Delete.

3.4: Delete the text and substitute: "The equipment shall have an upper test mean-time-between-failure (MTBF) (θ_0) and a lower test MTBF (θ_1) of 2000 hours and 1000 hours, respectively. (Upper test MTFB (θ_0) and lower test MTBF (θ_1) are as defined in MIL-STD-781.)".

3.6.3, line 3: Delete "Class IA" and substitute "Class A5".

3.6.5, first sentence: Delete "MIL-STD-167" and substitute "MIL-STD-167-1".

FSC 5825

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3.7.1, line 6: Delete "will be FHOF-3." and substitute "shall be Type FHOF-3 in accordance with MIL-C-915/6."

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3.7.2.2, line 8: Delete: "35A-7" and substitute "3SA-7".

3.8.1(a), line 5: After "TS-3389" Add "in accordance with MIL-T-28820".

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3.8.2, line 1: After "as" add "specified".

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3.10.1.2(g) line 1: Delete "6.2.3" and substitute "6.3.3"; line 3: Delete "6.2.4" and substitute "6.3.4".

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After "3.10.8", add the following new paragraphs:

"3.11 Computer resources. Navy standard computer resources shall be used to implement system requirements. If this is not feasible, justification (including technical analysis, life-cycle cost comparison and potential schedule impact) shall be provided to the procuring activity prior to the use of any other computer resources. This includes computers, microcomputers, microprocessors, displays, peripherals, interfaces, and software.

"3.12 Leakage current. The leakage current of the equipment shall not exceed 5 milliamperes (mA).

"3.13 Workmanship.

"3.13.1 General workmanship. Workmanship shall conform to Requirement 9 of MIL-STD-454. Soldering shall conform to Requirement 5 of MIL-STD-454.

"3.13.2 Workmanship screen. All equipments shall withstand a defect detection vibration screen of random type vibration at $0.04g^2$/Hz ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.6.11.2."

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4.4.1, line 7: After last sentence: Add: "The general workmanship and workmanship screen inspections shall be the first inspections conducted, and shall be performed in the order indicated in TABLE IV."

TABLE IV: Add the following new entries:

"General workmanship	3.13.1
Workmanship screen	
Vibration	3.13.2
Temperature cycling	3.13.2

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TABLE V: Delete "Heat test" and entry in its entirety.

TABLE VI, after "Inclination", Add:

"Leakage current 3.12".

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4.5 through 4.5.2: Delete and substitute:

"4.5 Reliability testing.

4.5.1 Reliability qualification testing. Reliability qualification testing shall be performed on at least two production equipments. This testing shall be accomplished in accordance with MIL-STD-781, using Test Plan IIIC and Category 3A test conditions. The equipments shall be operated on a continuous operating cycle during the test.

4.5.2 Production reliability acceptance testing. Every production equipment, with the exception of those subjected to reliability qualification testing, shall be subjected to a production reliability acceptance test, in accordance with the Accept-reject criteria figure of MIL-STD-781. The test duration for each equipment shall be 50 hours. During the test, the equipment shall be operated continuously in an ambient temperature of 40° Celsius (C) ±5°C. If a failure (see 6.3.6) occurs during the test, the equipment shall be repaired and another 50-hour test run to verify repair."

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After "4.6.8", Add the following new paragraphs and FIGURES:

"4.6.9 Leakage current. Leakage current shall be measured at the steady-state power line voltage and frequency, 115 volts (V), 60 Hz and 400 Hz. Leakage current shall be measured on one equipment of the first month's production. Thereafter, leakage current shall be measured on one equipment per 100 successive equipments produced.

WARNING

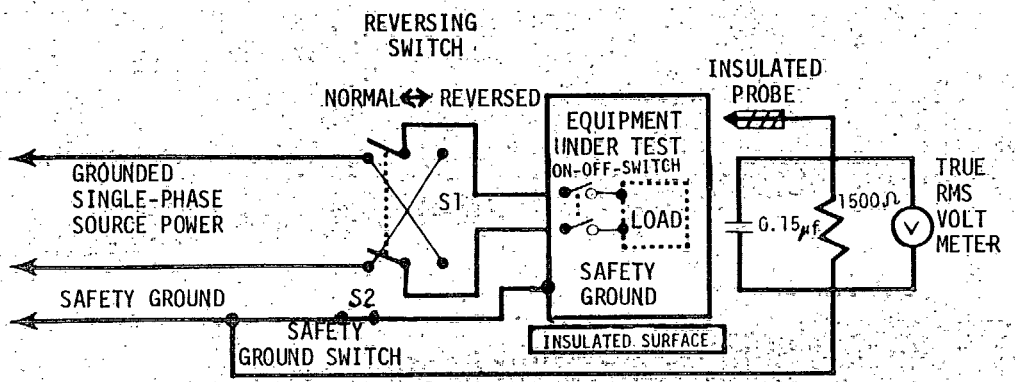
THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.6.9.1 Equipment test connections. After power removal, each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment and the source power shall be opened during the test. OBSERVE WARNING STATEMENT. The equipment shall be connected as shown in FIGURE 3 if it is connected to single phase source power.

4.6.9.2 Measurement. Leakage current shall be measured on the equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 V, represents 5 mA of leakage current. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housing, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 3. The open safety ground conductor shall be reconnected immediately after the test is completed."

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WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW REVERSED. ON-OFF SW ON. REPEAT STEP 4
7. ON-OFF SW OFF. REPEAT STEP 4.
8. S2 SW CLOSED. S1 SW NORMAL.
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

FIGURE 3. Single-phase test diagram for leakage current measurement.

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"4.6.10 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level, for conformance to the requirement of 3.13.1.

4.6.11 Workmanship screen. Prior to conducting temperature cycling, vibration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.6.11.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 4. The control accelerometer shall be located next to one of the mounting points of the item under vibration. Spectral analysis equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

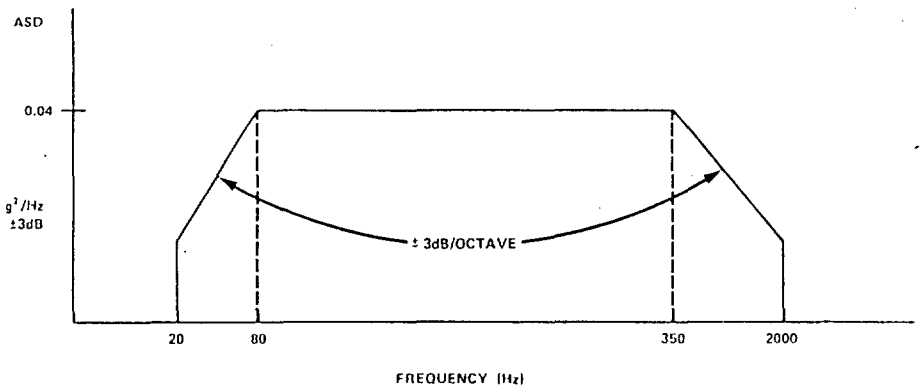


FIGURE 4. Random vibration curve.

4.6.11.2 Temperature cycling. Each equipment shall be subjected to 10 cycles of the temperature curve shown in FIGURE 5. The temperature rate of change shall be not less than 5°C per minute. The equipment power shall be turned on and off at the indicated times. The equipment shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the equipment reaches 80 percent of the chamber temperature. When failures occur, the equipment shall be reworked and the cycling continued for a cumulative total of 10 cycles.

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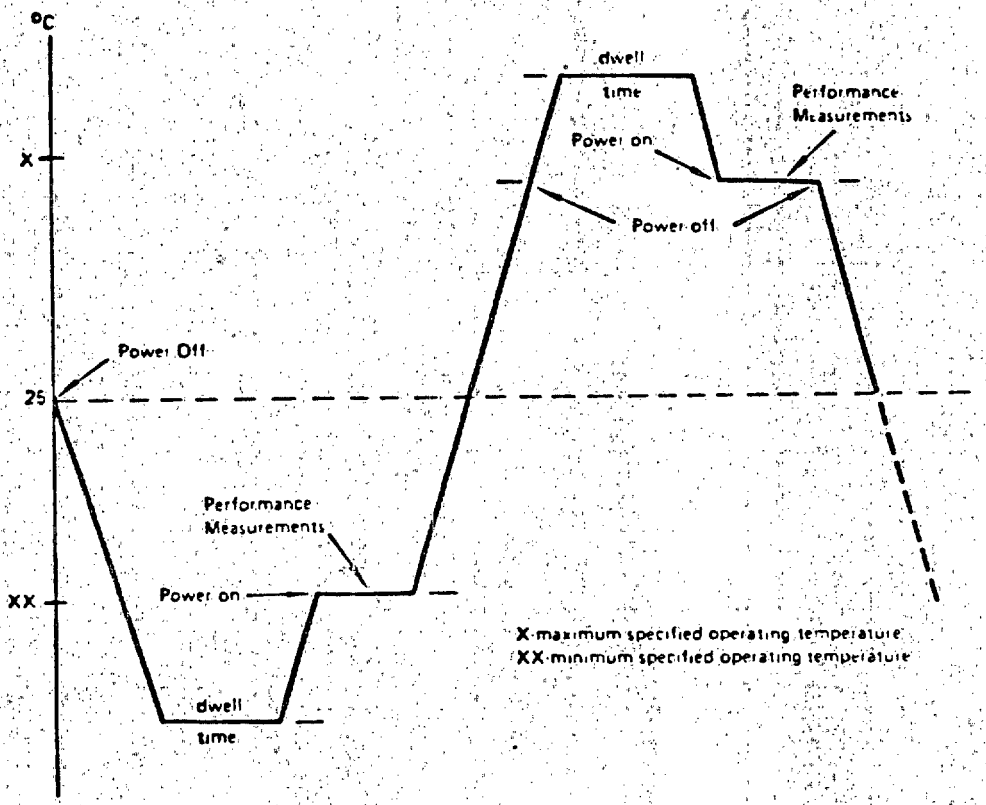


FIGURE 5. One cycle of temperature curve (not to scale)."

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